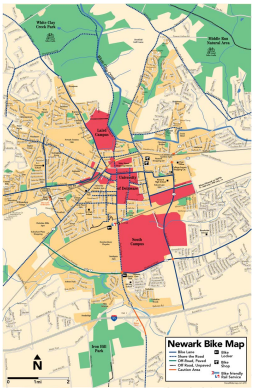
Slime mold (*Physarum polycephalum*): A paradigm for self-assembly of robust networks.



Background and motivation

Resource distribution networks play a crucial role at all scales of biological and artificial systems. Within an individual organism, cells receive nutrients and expel waste. Nutrients must be transported from a source to all cells efficiently. Similarly, waste must be collected from every cell and transported out of the organism. Communities of organisms and entire ecosystems have the same requirements and natural selection drives the system to create efficient *resource distribution networks.*

Artificial resource distribution networks such as our transportation system or our utility grid, rely upon a *top-down* design paradigm where a designer or team of designers lay out the network using the best analytic methods available. Sometimes these methods are rudimentary or even irrational. Modifications are made in an ad-hoc manner as needed. For instance, if one is building a new housing development, one builds a road connecting to the existing network. Unfortunately, a top-down management approach to a complex system which is not understood becomes meaningless. In recent years, there has been a drive to design and implement *biologically-inspired* systems.

Resource distribution networks are nothing new in the natural world. One encouraging observation is that very simple life forms are capable of developing efficient and robust resource distribution networks. *Physarum polycephalum* is one such organism. *Physarum polycephalum* is a single-celled organism that we can grow in colonies in the MEC Lab. The colony requires only nutrients (oat flakes), moisture, warmth and air to thrive. Moisture is supplied by the agar medium on the petry dish. Warmth and air are controlled at a constant value in the lab. The controllable variable is the quantity and location of the food. These single-celled creatures decide to form distribution tubes.

Central issues

Understanding slime mold requires one to answer a number of questions, beyond merely simulating a complex assembly of simple units.

1. What mathematical structures can be used to describe the slime mold colony? What are their strengths and weaknesses?

2. What is a mathematically precise definition of efficiency in a resource distribution network?

3. What is a mathematically precise definition of robustness in a resource distribution network?

4. How can a simple system of locally interacting cells produce an efficient network? What are the parameters and how to they affect efficiency and robustness?

5. Is there a selection mechanism that drives the colony toward efficiency or toward robustness? What is it?

Experiments

Physarum Polycephalum is easy to grow in the MEC Lab. Typically, an experiment takes two to three days. The MEC Lab has a digital SLR camera to record the state of the mold, and computer equipment for analyzing the colony. Matlab's image processing toolkit will be particularly helpful in measuring and analyzing the network. There is also a digital microscope for more detailed analysis.

Challenges

- Milestone #1: Find the most relevant and informative reference for your project and defend your determination.

- Milestone #2: You develop a model for the generation of the resource distribution network. It can a local model or a global model. I will create a colony and place food sources at certain points. Use your model to predict the resulting colony network.

- Milestone #3: You develop a local model for the generation of the resource distribution network. I will create a colony and place food sources at certain points. Use your model to predict the resulting colony network.

- Milestone #4: You design an experiment (placement of food sources) for another team. We will compare the performance of your model with their model.

- Milestone #5: I will create a colony and place food sources at certain points. Then, after the colony is established, I will remove one of the food sources. Use your model to predict the resulting colony network.

Literature

The most provocative experimental work on slime mold has been produced by an investigator named Nakagaki and his collaborators. Nakagaki's group make considerable claims about robustness and intelligence in the colonies. Make sure you view these claims critically. There is a group in the UK actively working this in area. I can share this work with you, but it cannot be distributed on the web. Network dynamics is a very trendy topic. For a quick overview, I refer you to Strogatz's paper.

References

[1] T. Nakagaki. Smart behavior of true slime mold in a labyrinth. *RESEARCH IN MICROBIOLOGY*, 152(9):767–770, 2001.

[2] T. Nakagaki, R. Kobayashi, Y. Nishiura, and T. Ueda. Obtaining multiple separate food sources: behavioural intelligence in the physarum plasmodium. *PROCEEDINGS OF THE ROYAL SOCIETY OF LONDON SERIES B-BIOLOGICAL SCIENCES*, 271(1554):2305–2310, 2004.

[3] T. Nakagaki, H. Yamada, and M. Hara. Smart network solutions in an amoeboid organism. *BIOPHYSICAL CHEMISTRY*, 107(1):1–5, 2004.

[4] T. Nakagaki, H. Yamada, and A. Toth. Maze-solving by an amoeboid organism. *NATURE*, 407(6803):470–470, 2000.

[5] T. Nakagaki, H. Yamada, and A. Toth. Path finding by tube morphogenesis in an amoeboid organism. *BIOPHYSICAL CHEMISTRY*, 92(1-2):47–52, 2001.

[6] S. H. Strogatz. Exploring complex networks. *Nature*, 410:268–276, 2001.

[7] A. Tero, R. Kobayashi, and T. Nakagaki. Physarum solver: A biologically inspired method of road-network navigation. *PHYSICA A-STATISTICAL MECHANICS AND ITS APPLICATIONS*, 363(1):115–119, 2006.

[8] P. Xu, B. Yu, M. Yun, and M. Zou. Heat conduction in fractal tree-like branched networks. *INTERNATIONAL JOURNAL OF HEAT AND MASS TRANSFER*, 49(19-20):3746–3751, 2006.